How sensitive are finite RH criteria? Certified Davenport–Heilbronn controls

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Abstract

Finite positivity tests related to the Riemann Hypothesis require calibration against functions with known off-line zeros. We study Davenport–Heilbronn character combinations and two Epstein zeta functions. An interval-certified Davenport–Heilbronn atlas through height 340 contains 235 critical-line zeros and seven reflected off-line pairs. A global angular comparison and Schoenberg's sector theorem give PF_{435} for the ordinary folded Taylor coefficients. All 18,291 tested solid minors at ranks 430–520 and shifts 0–200 are positive. A targeted calculation finds negative shift-44 bands at ranks 664–1099 and 1536–1972. The second band meets a forecast recorded before the new computation. Nearby shifts 40–48 reveal no earlier failure in the tested strip. A classical split-pair asymptotic explains infinite sign changes, while a dominant-pole argument gives eventual positivity at shift one. We also certify negative geometric-form witnesses for seven controls and test a recorded conductor-17 forecast. A rational mixture path decreases the lowest off-line offset from 0.03 to 0.004; its common-degree normalized support upper bound rises from 0.5699 to 0.6597. Larger-degree checks show only small endpoint changes. This finite path does not determine the small-offset limit. All window endpoints are support upper bounds; positive finite sections give no full-window lower bounds. Finite-atlas attribution has no negligible infinite-tail bound. These results establish neither a universal zero-free region nor the Riemann Hypothesis.

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1 Introduction

An all-orders positivity criterion can characterize the Riemann Hypothesis without making a finite instance a sensitive detector of an off-line zero. The distinction matters when choosing a finite computational target. Davenport–Heilbronn functions supply controls with a functional equation and zeros off the symmetry line. Their classical theory and numerical zero computations are established background [2, 3, 4]. All seven off-line centers in Table 1 were already reported: the first four by Spira and the final three by Balanzario–Sánchez-Ortiz, as recorded in [3]. Our contribution is the complete interval count and global angular comparison, rather than discovery of these zero locations.

Our main result is a reproducible calibration for one specified family of Toeplitz minors. We certify the steepest folded Davenport–Heilbronn zero and the resulting PF_{435} sufficient guarantee, then test the requested grid beyond the isolated quadratic's first negative minor. Every full-function

minor in that grid is positive. This is a failure of sharpness for the specified solid grid. A subsequent targeted shift-44 calculation finds a negative minor at rank 664. This is the first negative member of that column and the first failure among the explicitly listed DH contiguous minors tested in this study. It is not globally first and does not prove PF_{520}.

The window experiments address a different finite criterion. We construct rational trial functions with negative Weil-type quadratic form directly from the geometric side. The existing `teal-sea/zeta-lab` report [8] already found the classical Davenport–Heilbronn negative window near $C = 31$; that observation is prior work. We refine the finite-family search, add other controls, and distinguish the resulting upper bounds from a theorem about the minimum negative support. Bombieri's fixed-support spectral analysis [5] is direct prior art for this finite-to-infinite distinction.

A separate reproducibility check concerns finite zeta windows rather than these negative controls. We replayed Liu's frozen release [7], commit `b6cd218`: Theorem A at $L = 1$ passed retained-factor verification with fresh source, matrix and all-row residual reconstruction; Theorem B at $L = 17/16$ passed a clean rebuild and a second verification pass. All frozen source remained unchanged. A byte-identical private standard-library zip adapted the original Windows-oriented import allowlist to macOS; no acceptance gate was bypassed. This is computational re-execution of the supplied implementation, rather than an independent arithmetic backend or human peer review. Full receipts and the runtime adaptation are in the supplement. It is a fixed-window result and does not supply an all-support estimate.

The contributions of this preprint are:

- a complete interval-certified Davenport–Heilbronn atlas through height 340 and a global steepest-angle reduction;
- a fully resolved 18,291-cell solid-minor grid, with a second determinant identity checking selected high-rank cells;
- a common-degree seven-control window calibration, a recorded conductor-17 prediction test, and enclosed finite-atlas quartet contributions;
- an elementary fixed-shift masking lemma and a direct eventual-positivity argument for the classical Davenport–Heilbronn shift-one ladder;
- a split-pair asymptotic argument with nonzero leading coefficient and a targeted negative shift-44 minor at rank 664, followed by a second-band forecast test;
- three further complete small-offset mixture atlases and negative witnesses, with a common-degree collision-path calibration.

These are calibration results. The analytic reductions use classical functional equations, factorization and sector theorems; the numerical enclosures use Arb. An independent AI-agent review checked the split-pair argument and selected numerical certificates. Independent external human review remains pending.

2 Normalizations and the finite tests

2.1 Character combinations and Epstein controls

Let χ be a primitive odd nonreal character of conductor q, and put

$$\epsilon_\chi = \frac{\tau(\chi)}{i\sqrt{q}}, \qquad \kappa_\chi = \frac{\operatorname{Im}\epsilon_\chi}{1+\operatorname{Re}\epsilon_\chi}, \qquad H_\chi(s) = \frac{(1-i\kappa_\chi)L(s,\chi)+(1+i\kappa_\chi)L(s,\bar\chi)}{2}. \tag{1}$$

We exclude the singular normalization $1 + \operatorname{Re}\epsilon_\chi = 0$. The Dirichlet coefficients are real, periodic, and have first coefficient one. The completion

$$\Lambda_\chi(s) = (q/\pi)^{s/2}\Gamma((s+1)/2)H_\chi(s) \tag{2}$$

is entire and satisfies $\Lambda_\chi(s) = \Lambda_\chi(1-s)$. The classical control has $q = 5$, $\chi(2) = i$, and period $(0, 1, \kappa, -\kappa, -1)$, where

$$\kappa = \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}. \tag{3}$$

We also use $H_d(s) = E(s, x^2 + dy^2)/2$, $d = 5, 6$, normalized by counting representations. The class-number-two decompositions are

$$H_5(s) = \tfrac{1}{2}\{\zeta(s)L(s, \chi_{-20}) + L(s, \chi_{-4})L(s, \chi_5)\}, \tag{4}$$

$$H_6(s) = \tfrac{1}{2}\{\zeta(s)L(s, \chi_{-24}) + L(s, \chi_{-3})L(s, \chi_8)\}. \tag{5}$$

Here χ_D is the primitive quadratic character of fundamental discriminant D. The completion has gamma factor $(\sqrt{d}/\pi)^s\Gamma(s)$ and poles at 0 and 1. The entire symmetric target is therefore $s(s-1)(\sqrt{d}/\pi)^s\Gamma(s)H_d(s)$. The contour computation uses $F_d(s) = (s-1)H_d(s)$, removing the pole at 1; the prefactor relating it to the entire target introduces no additional zero inside the contours used here. Genus decompositions and their normalizations are checked against independent representation counts. The background for these controls is discussed in [4].

2.2 Ordinary coefficients and solid minors

For the classical completion define the folded entire function

$$\frac{\Lambda_5(1/2+z)}{\Lambda_5(1/2)} = F(z^2), \qquad F(w) = \sum_{n\ge 0} \mu_n w^n, \qquad \mu_0 = 1. \tag{6}$$

All Toeplitz tests use the ordinary coefficients μ_n. The classical Jensen coefficients $n!\mu_n$ are different data and are not substituted into these matrices. With $\mu_n = 0$ for $n < 0$, write

$$D_{r,s} = \det(\mu_{s+i-j})_{0\le i,j<r}, \qquad D_{0,s} = D_{r,0} = 1. \tag{7}$$

PF_m means nonnegativity of all Toeplitz minors of order at most m, including arbitrary row and column sets. Our solid grid is only a subset.

2.3 Quadratic forms and support

We use physical support $[-S/2, S/2]$, with

$$S = \log C. \tag{8}$$

For a real trial function let $\mathcal{F}_f(z) = \int_{-S/2}^{S/2} f(x)e^{zx}\,dx$. The spectral form associated with a real symmetric completion is

$$Q_H(f) = \sum_\rho \mathcal{F}_f(\rho - \tfrac{1}{2})\overline{\mathcal{F}_f(\tfrac{1}{2} - \bar{\rho})}. \tag{9}$$

Its geometric form uses the coefficients in $-H'/H = \sum_{n\ge 2} c_n n^{-s}$. For a character combination these coefficients need not have prime-power support or one sign. All active integers $n < C$ are

included. The odd gamma factor and conductor term in (2) are used, and there is no pole block. The Epstein forms have their own gamma and pole terms. The gamma duplication identity is used before reusing entry formulas.

The finite trial family consists of normalized cosines through order N in the even sector and normalized sines through order N in the odd sector, together with the constant even mode. Each sector is orthonormal in $L^2[-S/2, S/2]$. Stored coordinates use the shifted exponentials $S^{-1/2}e^{2\pi in(x+S/2)/S}$, so the real centered sine and cosine modes include the phase $(-1)^n$. This phase is retained in zero-side evaluation. An interval LDL factorization proposes a rational vector at its first negative leading pivot. The full geometric Rayleigh quotient of that exact vector is then enclosed independently of the pivot sign. A negative quotient proves a negative full-form trial. Positivity of the finite matrix proves only positivity on that finite space. The spaces obtained by changing S are not nested, so the searches below do not provide monotone brackets for the first negative full support.

3 The complete classical Davenport–Heilbronn atlas

Lemma 3.1 (Positive theta kernel)**.** *The classical completion is positive on the real axis. Every ordinary folded coefficient μ_n is strictly positive.*

Proof. Set $\theta(t) = \sum_{n\geq 1} na_n e^{-\pi n^2 t/5}$. Poisson summation for the odd character combination gives $\theta(t) = t^{-3/2}\theta(1/t)$. For $t \geq 1$, put $v = \pi t/5 > 1/2$. Since $a_2 = \kappa > 0$ and $|a_n| \leq 1$,

$$\theta(t) \geq e^{-v} - \sum_{n\geq 3} ne^{-vn^2} \geq e^{-v} - \frac{e^{-4v}}{2v} > 0. \tag{10}$$

The middle inequality uses the decreasing function xe^{-vx^2} on $[2, \infty)$. Consequently $\Psi(u) = e^{3u/2}\theta(e^{2u})$ is even and positive. Mellin inversion gives

$$\Lambda_5(1/2 + z) = 4\sqrt{\pi/5} \int_0^\infty \Psi(u) \cosh(zu)\, du. \tag{11}$$

The kernel decays faster than any exponential. This proves real-axis positivity and positivity of every even Taylor moment. □

Theorem 3.2 (Computer-assisted atlas and steepest angle)**.** *The completed classical Davenport–Heilbronn function has precisely 249 zeros, counted with multiplicity, in the upper half-plane through height 340. They consist of 235 critical-line zeros and seven reflected off-line pairs. All are simple. Every completed zero satisfies $|\operatorname{Re}\rho - \frac{1}{2}| < 1$. The zero near $0.8085171825 + 85.69934849i$ has the largest folded angular deviation among all completed zeros.*

Proof with explicit numerical inputs. The first coefficient of H_5 is one, and Arb gives

$$\sum_{n\geq 2} |a_n| n^{-3/2} = 0.7375906494\ldots < 1. \tag{12}$$

The sum decreases with the real part. Thus H_5 has no zero at $\operatorname{Re} s \geq 3/2$; the completed functional equation gives the reflected exclusion at $\operatorname{Re} s \leq -1/2$. There are no gamma poles inside this strip.

On the rectangle $[-1/2, 3/2] + i[0, 340]$, the real bottom has zero argument increment by Lemma 3.1; its gamma factor is positive there. The other three sides are covered by 5,208 Taylor image rectangles excluding zero. A convex zero-excluding image gives a continuous argument increment of magnitude less than π, so principal endpoint increments may be summed. Their total

encloses $2\pi \cdot 249$. There are 235 disjoint critical-line sign brackets and seven disjoint simple right-hand Rouché disks. Functional-equation reflection supplies seven left-hand disks. Their multiplicity count exhausts 249, proving completeness and simplicity. The atlas is replayed at higher precision with additional subdivision where needed.

For $\rho = 1/2 + b + i\gamma$, the zero of F at $w = (\rho - 1/2)^2$ has angular deviation

$$\phi_\rho = 2 \arctan \frac{|b|}{|\gamma|} \tag{13}$$

from the negative axis. The height-85.6993 disk has $\phi = 0.0071999578790\ldots$, greater than the deviations of all other off-line disks and greater than $2\arctan(1/340)$. The strip exclusion therefore controls every higher zero as well. Every comparison uses enclosing balls, rather than the rounded values in Table 1. □

Table 1: Right-hand classical Davenport–Heilbronn zeros below height 340. The reflected zeros and 235 critical-line zeros exhaust the contour count. Rounded coordinates are not certificate inputs.

$\operatorname{Re}\rho$	$\operatorname{Im}\rho$
0.8085171825	85.69934849
0.6508300806	114.1633427
0.5743560505	166.4793059
0.7242576946	176.7024612
0.8695305796	240.4046724
0.8195495921	320.8764897
0.7682231236	331.0502594

4 The sector guarantee and the high-rank grid

Theorem 4.1 (Sufficient sector guarantee)**.** *The ordinary coefficient sequence in* (6) *belongs to* PF_{435}.

Proof. The folded function has order 1/2, hence a genus-zero product with constant one. By Theorem 3.2 every zero lies in the wedge about the negative axis of half-angle ϕ, where interval calculation gives

$$436 < \frac{\pi}{\phi} = 436.3348656172\ldots < 437. \tag{14}$$

Schoenberg's sufficient sector theorem, stated as Theorem B in [6], gives PF_{435} for every finite conjugation-invariant product prefix: its wedge half-angle is less than $\pi/436$. The prefixes converge locally uniformly. Each fixed Toeplitz minor is a polynomial in finitely many coefficients, so nonnegativity passes to the entire function. □

For the isolated quadratic $A(w) = 1 + 2\cos\phi\, w + w^2$, expansion of the tridiagonal determinant gives

$$D_{r,1}(A) = U_r(\cos\phi) = \frac{\sin((r+1)\phi)}{\sin\phi}. \tag{15}$$

Equation (14) makes rank 436 its first negative shift-one minor. The angular rank scale is $\pi|\gamma|/(2|b|)$, retaining the offset b.

Proposition 4.2 (Certified solid grid)**.** *For the full folded Davenport–Heilbronn function,* $D_{r,s} > 0$ *for every* $430 \le r \le 520$ *and* $0 \le s \le 200$*.*

Computational verification. We generate 721 ordinary coefficients through index 720 at 32,768 bits. The positive rescaling $\mu_n \mapsto 100^n \mu_n$ preserves determinant signs. Desnanot–Jacobi condensation evaluates

$$D_{r,s} = \frac{D_{r-1,s}^2 - D_{r-1,s-1} D_{r-1,s+1}}{D_{r-2,s}}. \tag{16}$$

Every divisor is required to exclude zero; unresolved entries invalidate their descendants. All 18,291 requested cells resolve positive. This count includes the 91 trivial shift-zero cells. The full sign digest is replayed. For $h_n = [w^n]1/F(-w)$, Jacobi's complementary-minor identity also gives

$$D_{r,s} = \det(h_{r+i-j})_{0 \le i,j < s}. \tag{17}$$

It independently checks 24 high-rank cells at ranks 430, 435, 436, 437, 480 and 520, and shifts 1, 2, 5 and 20. These checks share the interval coefficient input and FLINT backend, while bypassing the condensation divisions. □

Figure 2 displays the distinction between the isolated factor and the full-function grid. The sufficient sector bound is not sharp for this tested family. Arbitrary row and column sets and larger shifts remain untested, so Proposition 4.2 supplies no global first-failure rank or PF_{520} claim.

5 Why multiplication can conceal a negative factor

Toeplitz multiplication and Cauchy–Binet expand a product minor into a sum of products of factor minors. One negative term need not determine the sum. The following elementary example makes that obstruction quantitative. It is included as an explanation, without a novelty claim.

Lemma 5.1 (A fixed-shift ladder can miss a conjugate pair forever)**.** *Let* $0 < c < 1$ *and*

$$G(w) = (1 + 4w)(1 + 2cw + w^2). \tag{18}$$

Then G *has two nonreal zeros, yet*

$$D_{r,1}(G) \ge \frac{2}{9}\, 4^r > 0 \qquad (r \ge 0). \tag{19}$$

Proof. The two nonreal zeros belong to the quadratic, while $1 + 4w$ adds a negative real zero. The reciprocal generating identity gives

$$D_{r,1}(G) = [w^r] \frac{1}{G(-w)} = \sum_{j=0}^{r} 4^{r-j} U_j(c). \tag{20}$$

Writing $c = \cos\phi$ with $0 < \phi < \pi/2$, the finite exponential sum for $U_j(\cos\phi)$ gives $|U_j(c)| \le j + 1$. Since $U_0 = 1$,

$$4^{-r} D_{r,1}(G) \ge 1 - \sum_{j=1}^{\infty} \frac{j+1}{4^j} = 2 - (1 - \tfrac{1}{4})^{-2} = \frac{2}{9}. \tag{21}$$

□

This lemma concerns one fixed-shift ladder. Other minors of G may fail. It does not prove the full Davenport–Heilbronn grid by factorization, since the remaining Davenport–Heilbronn factors include further off-line pairs. It explains why the first negative minor of an isolated quadratic is not an upper bound for the corresponding full-function ladder.

Proposition 5.2 (Eventual positivity of the DH shift-one ladder)**.** *Let γ_1 be the lowest positive Davenport–Heilbronn zero ordinate and put $a = \gamma_1^2$. There is a constant $A > 0$ such that*

$$D_{r,1}(F) = Aa^{-r} + O(64^{-r}) \qquad (r \longrightarrow \infty). \tag{22}$$

In particular, the full-function shift-one minors are strictly positive at every sufficiently large rank, despite the off-line zeros.

Proof. The complete atlas places the first two positive critical-line ordinates in $(81/16, 41/8)$ and $(143/16, 9)$, respectively. There is no off-line zero below height 85. Thus the first folded zero $-a$ is simple and unique in $|w| \le 64$, since $(41/8)^2 < 64 < (143/16)^2$. Write $B(w) = F(-w)$. We have $B(0) = 1$, and B is real and nonzero on $[0, a)$. Its simple first crossing therefore has $B'(a) < 0$. Set $A = -1/(aB'(a)) > 0$. The function

$$\frac{1}{B(w)} - \frac{A}{1 - w/a}$$

is analytic on a neighborhood of the closed disk $|w| \le 64$. Cauchy's coefficient estimate and $D_{r,1}(F) = [w^r]1/B(w)$ prove (22). □

This is ordinary dominant-pole analysis, included without a novelty claim. It does not give an explicit last exceptional rank, prove positivity at every rank, or control other shifts. It shows directly how a nearer real zero can dominate this particular DH ladder.

6 A split pair and the targeted failing column

The complementary determinant in (17) is the Jacobi–Trudi determinant for a rectangular Schur function of the inverse zeros. Its finite-product expansion is classical Widom–Schur theory; see Alexandersson [1], Theorem 1 and Section 3.2. For the entire function, the following residue argument states the additional analytic step explicitly. No novelty of this mechanism is claimed.

Theorem 6.1 (Split-pair oscillation)**.** *Let $s \ge 2$ and let B be real entire with $B(0) = 1$. Suppose its zeros of smallest modulus are $s - 1$ distinct simple zeros $a_1, \dots, a_{s-1}$, followed by exactly one simple nonreal conjugate pair $a, \bar{a}$. Suppose*

$$\max_j |a_j| < |a| < R,$$

and there are no further zeros in $|z| \le R$. Assume the inner set is conjugation invariant and $P = \prod_{j<s} a_j^{-1} > 0$. Write $h_n = [z^n]1/B(z)$, $\lambda_j = a_j^{-1}$, $\lambda = a^{-1} = |\lambda|e^{i\phi}$, with $0 < \phi < \pi$, and $d_r = \det(h_{r+i-j})_{0 \le i,j < s}$. Then for some $0 < \rho < 1$,

$$\frac{d_r}{(P|\lambda|)^r} = 2|K| \cos(r\phi + \arg K) + O(\rho^r), \qquad K \ne 0. \tag{23}$$

Consequently d_r is strictly positive infinitely often and strictly negative infinitely often.

Proof. For each of the $s+1$ stated poles put $c_j = -1/(a_j B'(a_j))$, using the same notation for the conjugate pair. Subtract their principal parts from $1/B$. Cauchy's estimate on $|z| = R$ gives

$$h_n = \sum_{j=1}^{s+1} c_j \lambda_j^n + O(R^{-n}).$$

Each pole contributes a rank-one matrix to the determinant. Cauchy–Binet gives the coefficient of $(\prod_{j\in I} \lambda_j)^r$, for an s-element pole set I, as

$$K_I = (-1)^{s(s-1)/2} \frac{\displaystyle\prod_{j\in I} c_j \prod_{\substack{i<j \\ i,j\in I}} (\lambda_j - \lambda_i)^2}{\displaystyle\prod_{j\in I} \lambda_j^{s-1}}. \tag{24}$$

Every factor is nonzero, since the poles are nonzero, distinct and simple. The two dominant sets contain all inner poles and one member of the pair; their constants are $K, \bar{K}$. Every other all-pole product is smaller in modulus by at least $|\lambda|/\min_{j<s} |\lambda_j| < 1$.

For completeness, expand the determinant by columns, choosing $k \geq 1$ remainder columns. The other $s-k$ rank-one columns must come from distinct poles; repeated choices vanish by alternation. Their largest possible product is the product of the $s-k$ largest inverse-pole moduli. After division by $(P|\lambda|)^r$, the resulting exponential rate is at most $(R|\lambda|)^{-kr}$. All column-offset factors and remainder constants are fixed because s is fixed. Thus one may take

$$\rho = \max\left\{ \frac{|\lambda|}{\min_{j<s} |\lambda_j|}, \frac{1}{R|\lambda|} \right\} < 1.$$

This proves (23).

If $\phi/(2\pi)$ is irrational, its rotation orbit is dense and visits arcs with cosine bounded above and below by nonzero constants. If it is rational, the orbit has at least three points; its cosine values sum to zero and cannot all vanish, so it also contains both signs. The decaying error therefore preserves both signs along infinite subsequences. □

Proposition 6.2 (The DH shift-44 failure)**.** *The classical folded DH function has $D_{r,44} \geq 0$ for $1 \leq r \leq 435$, $D_{r,44} > 0$ for $436 \leq r \leq 663$, and $D_{r,44} < 0$ for $664 \leq r \leq 1099$, and $D_{r,44} > 0$ for $1100 \leq r \leq 1400$. In particular, rank 664 is the first negative rank of this column. The column also takes both signs infinitely often.*

Proof and computer-assisted inputs. For the height-85.6993 zero, the modulus of the folded zero is $\gamma^2 + b^2$. The complete atlas encloses exactly 43 critical-line zeros below $\sqrt{\gamma^2 + b^2}$, with strict gaps on both sides. Every earlier folded zero is simple and negative real. All other off-line pairs and all zeros above the atlas lie farther away. Theorem 6.1 applies to $B(z) = F(-z)$, $s = 44$. The leading constant in (24) is nonzero by the simple disjoint root disks; its interval evaluation also excludes zero. Its phase is approximately 3.0787871477, and $2\pi/\phi = 872.6697312345\ldots$. These are asymptotic diagnostics, rather than finite-rank error bounds.

Theorem 4.1 covers ranks through 435. The preceding grid covers 436–520 at shift 44. For ranks 521–677, reciprocal coefficients are computed from the existing 32,768-bit ordinary coefficient enclosures. Condensation in the complementary determinant size through 44 has no unresolved divisor. It resolves every rank through 663 positive and 664–677 negative. Direct 44×44 determinants independently check ranks 520, 521, 663, 664 and 677 using the same coefficient input.

A fresh calculation through coefficient 1450 uses conjugate Dirichlet-L jets, separately from the earlier four-Hurwitz-jet evaluation, at the same 32,768-bit precision. All 721 shared coefficients overlap. With these 1,451 ordinary coefficient enclosures, the complementary calculation resolves every rank 521–1400: positive through 663, negative at 664–1099, and positive again at 1100–1400. A 33,024-bit determinant replay checks all 880 signs and direct 44×44 determinants at both band boundaries. The two coefficient paths and all determinant replays share FLINT/Arb; this is not an independent arithmetic backend. □

The first negative rank is specific to this column. Other shifts and non-solid minors may fail earlier, so no globally first-failure claim is made. The new calculations cover the fixed shift-44 column and the requested shifts 40–48 strip.

A recorded second-band forecast. Before computing the new cells, we recorded entry ranks 1535–1539 and last-negative ranks 1970–1974, using the previously measured scale $2\pi/\phi = 872.6697312345\ldots$. This was a local pre-registration, not an independently time-stamped public registration. The new interval calculation resolves the entire column through rank 2000: the second negative band is exactly 1536–1972, with positive ranks 1401–1535 and 1973–2000. The two endpoint errors relative to the nominal forecast $(1537, 1972)$ are $(-1, 0)$. No phase or frequency was refitted. The first-to-second band entry displacement is 872 ranks and the last-negative displacement is 873, consistent with the noninteger oscillation scale. This is a quantitative forecast check, not a uniform finite-rank remainder theorem.

The additional coefficients are enclosed without a new high-order jet. The positive-theta lemma gives $\mu_n \geq 0$, hence for every real $R > 0$,

$$0 \leq \mu_n \leq F(R)R^{-n}.$$

We retain the first 1,451 evaluated coefficient balls and append these bounds at $R = 2^{20}$ through index 2050. An outward enclosure of $F(R) = \Lambda(1/2 + \sqrt{R})/\Lambda(1/2)$ suffices for all 600 additional bounds. All second-band signs resolve with these broad enclosures. The written positive-coefficient lemma is an explicit analytic dependency; the tail intervals are not claimed to be fresh coefficient evaluations. A higher-precision replay and direct determinants check both band edges.

The requested neighboring strip contains 1,620 cells: shifts 40–48 and ranks 521–700. Every cell resolves; only shift 44 has negative members, starting at rank 664. Together with the historical ranks 430–520, shifts 0–200 grid and the targeted shift-44 column through rank 2000, rank 664 at shift 44 is the first failure among the DH contiguous minors *tested in this study*. Earlier ranks at other untested shifts and non-solid minors remain open. No claim about a globally first failure is made.

For a first isolated off-line pair, the split shift is the number of preceding folded zeros plus one. When a strict gap identifies that count with critical-line ordinates below $\sqrt{\gamma^2 + b^2}$, it is $s_* = N_{\text{line}}(\sqrt{\gamma^2 + b^2}) + 1$. The rank oscillation scale is $2\pi/\phi \sim \pi\gamma/|b|$. Height alone does not determine the first negative rank: the leading phase, residue amplitudes and finite-rank correction also matter.

7 Seven-control window calibration

We compare the finite-family negative endpoints at common degree $N = 192$ and 2,048-bit arithmetic. Table 2 reports the lowest off-line zero from each complete finite atlas, the positive finite endpoint, and the certified negative endpoint. Complete atlases reach height 160 for the additional

character combinations and 120 for the Epstein controls. A conductor-11 exploratory combination has no off-line zero below 160 and is excluded from the fitted seven-point sample.

The leading zero densities motivate a normalized resolution coordinate, rather than an exchange theorem:

$$T_*(C) = \begin{cases} 2\pi C/q, & \text{odd character completion of conductor } q, \\ \pi\sqrt{C/d}, & \text{Epstein form } x^2 + dy^2, \end{cases} \qquad r = \frac{T_*(C)}{\gamma}. \tag{25}$$

These expressions equate the leading ordinate density with the exponential type budget $S/(2\pi)$. That calculation does not prove that an off-line zero is detected at $T_*(C)$.

Table 2: Common-degree calibration, $N = 192$. C_+ means positive finite evidence; C_- is a certified negative full-form trial. The values are rounded; exact rational endpoints are retained in the supplement. $r = T_*(C_-)/\gamma$.

Control	γ	b	C_+	C_-	r
DH5	85.699348	0.308517	30.628746	30.630219	0.44914
DH13(2)	31.434205	0.083622	33.750244	33.751953	0.51896
Epstein5	15.668250	0.432970	19.825439	19.827148	0.39928
DH17(3)	105.155287	0.163306	133.738281	133.751953	0.47011
DH7(3)	31.580361	0.535699	13.008621	13.010651	0.36980
DH13(6)	25.608746	0.362174	21.099091	21.101334	0.39825
Epstein6	15.117579	0.303749	31.778137	31.781708	0.47828

Before the new conductor-17 window search, the prediction $C = 140$, with the interval $[126, 154]$, was recorded from the earlier three-control calibration. Its atlas proves that $0.6633061848 + 105.15528688i$ is the lowest off-line zero. The negative endpoint $C = 133.751953125$ is 4.46% below the forecast and falls inside the predicted interval. Degree 80 remains positive at every tested point through 200, whereas degrees 128 and 192 give negative trials at 140. Thus a detector-resolution check is necessary even for the empirical test. At degree 256, the tested section is positive at $C = 133.5$ and a rational witness is negative at $C = 133.73828125$. This modest change from degree 192 supports finite-family resolution stability at this control. None of these positive sections certifies a positive full window.

Figure 4 shows the seven finite-family upper-bound coordinates and descriptive fits. The fitted samples are not independent measurements of minimum support. The two Epstein observations offer only limited information on a degree effect, and the conductor-7 control has a zero to the right of 1. These distinctions limit extrapolation to zeta. The simple fit $r = a + kb$ has root-mean-square residual 0.0183. Adding a degree term reduces it to 0.0149, but the largest absolute leave-one-out error in the implied C rises from 16.3% to 19.6%. Seven support upper bounds therefore do not establish a universal 10%-accurate exchange law. In particular, the fitted intercept $r(0) \approx 0.54$ is an extrapolation into an untested regime. It does not establish bounded detection cost as an isolated off-line pair approaches the critical line. The offset limit requires separate controls and uniform cancellation and tail estimates; a linear continuation of the present fit is not assumed.

7.1 A held-out small-offset mixture

To probe a smaller offset while keeping the conductor and gamma factor fixed, we form the explicit rational mixture

$$H_t = (1-t)H_{13,2} + tH_{13,6}, \qquad t = \frac{30142204300589527246625322252726073}{10^{37}}. \tag{26}$$

The subscripts denote the conductor and Conrey label in (1). This is a mixture of two controls, rather than a single character-label control. Each summand has the same root-number-one functional equation and first Dirichlet coefficient one, so the mixture preserves these normalizations. We selected t to place a zero near $\operatorname{Re} s = 0.53$, then fixed the exact rational parameter before window trials. A complete atlas through height 160, independently replayed with 320-bit contours and 640-bit local disks, contains 113 critical-line zeros and five reflected off-line pairs, for 123 upper-half-plane zeros. The lowest off-line zero is $0.53 + 31.4336216845i$ to the displayed precision.

Before these window trials, the unchanged seven-control fit predicted $C = 34.5767528427$. At common degree 192, the section at $C = 37$ is positive, whereas an exact rational witness at $C = 593/16 = 37.0625$ has enclosed quotient approximately -7.37956×10^{-18}. The resulting $r = 0.5698713$ is 7.19% above the recorded forecast, within the earlier 16–20% leave-one-out errors. At degree 256, both $C = 37$ and $C = 37.0625$ already admit negative witnesses. This illustrates why a positive finite section cannot supply a full-window lower bound. The original seven-point regression is not refitted using this probe.

This experiment does not separate the proposed behaviours as $b \to 0$. One finite-offset support upper bound cannot establish or exclude unbounded detection cost, and the mixture retains other off-line pairs. The data show a modest increase in difficulty at the tested offset, rather than a limiting law. In its finite-atlas attribution, the lowest quartet contributes approximately -1.11965×10^{-13}, more than 100 times the magnitude of the next individual negative quartet. The finite total is approximately -1.15183×10^{-14}, substantially different from the geometric quotient. The omitted terms remain material; dominance within the atlas does not prove an isolated-pair detection theorem.

7.2 Following the collision path

We next fix three rational parameters targeting offsets 0.015, 0.008 and 0.004, using the same two conductor-13 components. Each complete atlas through height 160 again contains 113 line zeros and five reflected off-line pairs. All three atlases receive a separate higher-precision replay. The ordinates of their lowest off-line zeros stay near 31.43354. Their simple off-line disks have radius 10^{-80}, far below the smallest reflected-pair separation, which exceeds 0.0079. No unresolved collision or uncounted pair is passed over by a sampled zero search.

Table 3 compares exact negative-support upper bounds. The new trials use 4,096-bit arithmetic at common degree 192. Increasing the retained degree to 256 lowers each new negative endpoint by less than 0.05 in C, much less than the observed change across the path. These degree checks compare finite-family resolution; their positive sections remain evidence only.

Table 3: One conductor-13 mixture path. Offsets and normalized coordinates are rounded; the negative cutoffs are exact. C_{256} is another negative-support upper bound. At offset 0.03 it is the earlier single trial, rather than a newly bisected endpoint.

b	C_{192}	C_{256}	r_{192}
0.030	37.0625	37	0.569871
0.015	39.5625	39.515625	0.608312
0.008	41.28125	41.2421875	0.634740
0.004	42.90625	42.875	0.659726

The measured coordinate rises by 15.77% as the offset decreases by a factor of 7.5. A descriptive fit to the four degree-192 upper bounds is $C \approx 37.30 + 2.89 \log(0.03/b)$, with residuals recorded

in the supplement. This resembles logarithmic growth over the tested range; it does not establish divergence, a bounded limit, or an asymptotic rate for minimum negative support. All four points lie on one mixture path, other off-line pairs remain, and no full positive-support lower bound is available. The original seven-control regression is unchanged.

The degree-192 endpoint quotients for the three new controls are approximately -1.8864×10^{-19}, -2.1509×10^{-19}, and -1.9304×10^{-21}. Their sizes depend on the selected trial and endpoint; monotonicity in b is not asserted. Finite-atlas attribution identifies the lowest quartet as the largest individual negative contributor in each case. The finite totals still differ materially from the geometric quotients, so this is not an isolated-pair detection theorem.

8 Finite-zero-sum attribution

For a normalized real even witness, an off-line quartet represented by $z = b + i\gamma$ contributes

$$4\,\mathrm{Re}\{\mathcal{F}_f(z)^2\}. \tag{27}$$

For a real odd witness the contribution has the opposite sign. A critical-line pair contributes $2|\mathcal{F}_f(i\gamma)|^2$. We refine the atlas disks to radius 10^{-120}, verify containment in their parent disks or sign brackets, and evaluate these finite contributions with interval arithmetic.

Table 4 identifies the largest enclosed negative quartet contribution for each negative endpoint witness. The supplementary records retain the positive line sum separately from the finite total. This answers which quartet carries the negative term within the atlas. It does not prove that the omitted infinite tail is negligible at the scale of the geometric quotient. The exact geometric certificate proves negativity; the finite zero sum supplies mechanism evidence. A non-smooth Fourier witness does not automatically give the tail decay needed to promote that evidence to an independent full spectral certificate.

Table 4: Attribution for each common-degree negative endpoint witness. The dominant negative quartet is the lowest off-line pair in all seven finite atlases. P_H is the sum over the complete atlas below height H; Q is the certified geometric quotient. Their difference is material, so the finite sum is mechanism evidence rather than a negligible-tail spectral certificate.

Control	Parity	Dominant γ	Q	P_H
DH5	even	85.699348	-9.772×10^{-33}	-2.519×10^{-30}
DH13(2)	even	31.434205	-6.173×10^{-17}	-4.418×10^{-13}
Epstein5	even	15.668250	-1.031×10^{-8}	-1.296×10^{-4}
DH17(3)	even	105.155287	-1.483×10^{-43}	-5.005×10^{-40}
DH7(3)	odd	31.580361	-9.555×10^{-10}	-4.135×10^{-6}
DH13(6)	even	25.608746	-3.045×10^{-10}	-8.222×10^{-7}
Epstein6	even	15.117579	-3.789×10^{-9}	-2.180×10^{-5}

9 Reproduction and limits of inference

The code uses Python 3.12, python-flint 0.8.0, and Arb/FLINT interval arithmetic. Discovery Newton steps and floating-point pilot scans select candidates only. The accepted root disks, contour images, determinant signs, Rayleigh quotients, and frequency panels are checked with outward enclosures. The supplement records source hashes, input hashes, exact rational trial coefficients,

pivot intervals, and replay commands. Regenerable full matrix arrays are omitted from compact records; the initial generator is preserved by hash. The paper and compact supplement have a reserved archival identifier; publication is controlled by the author. External human analytic review remains pending.

The classical atlas uses `dh_zero_atlas.py` and `replay_dh_atlas.py`. The existing solid grid and complementary checks use `dh_high_rank_scan.py` and `replay_dh_high_rank.py`. The subsequent targeted column and nearby strip are recorded separately. New control atlases use `character_zero_atlas.py` and `epstein_zero_atlas.py`. The geometric calibration and finite attribution use `control_window_calibration.py` and `control_zero_attribution.py`. The targeted column uses `dh_split_pair_column.py`; `dh_split_pair_asymptotic.py` checks its modulus ordering and leading constant, and `dh_split_pair_coefficients.py` supplies the separate Dirichlet-L coefficient evaluation. The full targeted record is `column44_rank521-1400_32768bits.json`, with `column44_extended_replay.json` from `replay_dh_split_pair_extended.py`. The small-offset mixture uses `near_line_mixture_control.py`, `near_line_window_calibration.py` and `near_line_attribution.py`; its exact definition, complete atlas, recorded forecast and window records are retained separately from the seven-control fit. The second-band forecast, coefficient bounds and nearby strip use `assess_second_band_forecast.py`, `dh_cauchy_coefficient_extension.py` and `dh_targeted_followup_scan.py`. The collision path uses `near_line_mixture_path.py`, `mixture_path_resolution.py` and `attribute_mixture_path.py`. The current supplement is checked with `verify_second_band_round.py`; older full-paper manifests bind historical manuscript revisions. The initial column record is `column44_rank521-677_32768bits.json`; its coefficient input is the existing `dh_rank430-520_shift0-200_32768bits_coefficients.json`.

The finite Schur inertia identity after completion of the square and the actual quadratic Toeplitz determinant formula are checked in Lean 4. Their six-theorem audit uses only `propext`, `Classical.choice`, and `Quot.sound`. The analytic explicit formulas, entire-function factorization, Schoenberg sector theorem, and interval computations remain separate written and computer-assisted dependencies.

The data support testing a calibrated detector. They do not establish *positivity at C excludes every off-line zero below approximately* $12C$. Such an implication additionally requires a uniform theorem over offsets, multiplicities, local zero configurations, and the infinite tail. The mixture path reaches offset 0.004, but its four finite support upper bounds do not determine the small-offset limit. Likewise, the solid-grid result does not characterize all minors of a given rank. These controls provide no independent confirmation of a new zeta window-positivity record and no route to RH by themselves.

10 Discussion

Two finite criteria show different sensitivities. The Toeplitz grid remains positive after the isolated factor becomes negative, and Lemma 5.1 demonstrates an exact fixed-shift masking mechanism. Proposition 5.2 proves eventual positivity for the classical Davenport–Heilbronn shift-one ladder itself. At shift 44, the first negative member occurs at rank 664, and the split-pair theorem gives infinite sign changes. Thus the shift, folded-modulus ordering, phase and rank all matter. The window family produces negative rational witnesses at reproducible cutoffs, with a conductor-17 negative endpoint inside its forecast range. The small-offset path shows measured growth with stable endpoints across the two retained degrees, without deciding a limiting law. The shared feature is that resolution and the zero configuration must be accounted for before interpreting a

finite sign.

The near-term mathematical target is a quantitative detector with a proved tail and cancellation bound. The near-term computational target is a certificate whose full comparison cost is measured, including the infinite complement. Calibration of either target requires controls such as those used here. Neither replaces the missing all-support arithmetic estimate.

Assistance and review. AI assistance was used for code, calculations and drafting. A separate AI agent reviewed the split-pair argument, replayed the targeted initial column at higher precision and reassembled the small-offset negative witness. This is not human peer review; the author remains responsible for the manuscript. External human review of the atlas/sector and split-pair arguments is required before the author-controlled publication step.

A Entry formulas and removable-point contour images

Here are explicit formulas for regenerating the new finite matrices. Put $\omega_m = 2\pi m/S$, $b_k = 2k+2\alpha$, where $\alpha = 1/4$ or $3/4$. The no-pole gamma entry is

$$\begin{aligned} G_\alpha(m,m) = & -\log\pi + \operatorname{Re}\psi(\alpha - i\omega_m/2) + \frac{\operatorname{Re}\psi'(\alpha - i\omega_m/2)}{2S} \\ & - \frac{2}{S}\sum_{k\ge 0} e^{-b_k S}\frac{b_k^2 - \omega_m^2}{(b_k^2+\omega_m^2)^2}, \end{aligned} \tag{28}$$

$$\begin{aligned} G_\alpha(m,n) = & -\frac{\operatorname{Im}\psi(\alpha - i\omega_m/2) - \operatorname{Im}\psi(\alpha - i\omega_n/2)}{2\pi(m-n)} \\ & - \frac{1}{\pi(m-n)}\sum_{k\ge 0} e^{-b_k S}\left(\frac{\omega_m}{b_k^2+\omega_m^2} - \frac{\omega_n}{b_k^2+\omega_n^2}\right) \quad (m \ne n). \end{aligned} \tag{29}$$

They follow by expanding the gamma kernel and integrating the shifted Fourier correlations. Truncation before $k = K$ has a uniform absolute error

$$\varepsilon_{\alpha,K} \le \frac{2e^{-(2K+2\alpha)S}}{S(2K+2\alpha)^2(1-e^{-2S})}. \tag{30}$$

For the character controls use $G_{3/4} + \log(q)I$ and subtract the finite log-derivative correlation sum. For the Epstein controls use $G_{1/4} + G_{3/4} + \log(4d)I$, the one pole block, and that control's log-derivative correlation sum. Selected entries are checked against adaptive integration of the original regularized geometric kernel. The parity matrices also obey the exact divided-difference formulas implemented in the generator.

For completeness, the genus identities (4)–(5) can be obtained from the two reduced ideal classes. Their forms are $(1,0,5),(2,2,3)$ at discriminant -20, and $(1,0,6),(2,0,3)$ at discriminant -24. Decompose the principal partial zeta function into the trivial class character and the nontrivial genus character. The latter Euler products factor as $L(\chi_{-4})L(\chi_5)$ and $L(\chi_{-3})L(\chi_8)$, respectively. The unit group has order two, giving the representation normalization $E/2$.

The additional contours enclose all four sides, including the real bottom; they do not assume the classical DH positive-theta lemma for another control. FLINT's direct series evaluations on nonzero-radius cells containing removable points can be indeterminate. We use analytic Taylor images there. For a character combination with zero period sum, partial summation gives $|H(s)| \le$

$|s|\sum_{n<q}|a_n|/\operatorname{Re} s$ for $\operatorname{Re} s > 0$. Thus $M = 3\sum_{n<q}|a_n|$ bounds $|H|$ on $|s-1| \le 1/2$. The Taylor remainder through order K on $|s-1| \le r < 1/2$ is at most

$$M\,\frac{(2r)^{K+1}}{1-2r}. \tag{31}$$

For $F_d = (s-1)H_d$, fractional-part partial summation gives $|(s-1)\zeta(s)| \le 3$ on the same disk. If M_j are the three character-period absolute sums in the genus decomposition, a valid bound is $(9M_0 + (9/2)M_1M_2)/2$. Near zero the regular functional-equation identity is

$$F_d(s) = \frac{\Gamma(2-s)}{\Gamma(1+s)}(\sqrt{d}/\pi)^{1-2s}F_d(1-s). \tag{32}$$

Cauchy bounds on a slightly larger Taylor disk also enclose the required derivative coefficients. These images, ordinary Taylor images elsewhere, and the Rouché inequalities are the numerical inputs to the finite zero counts.

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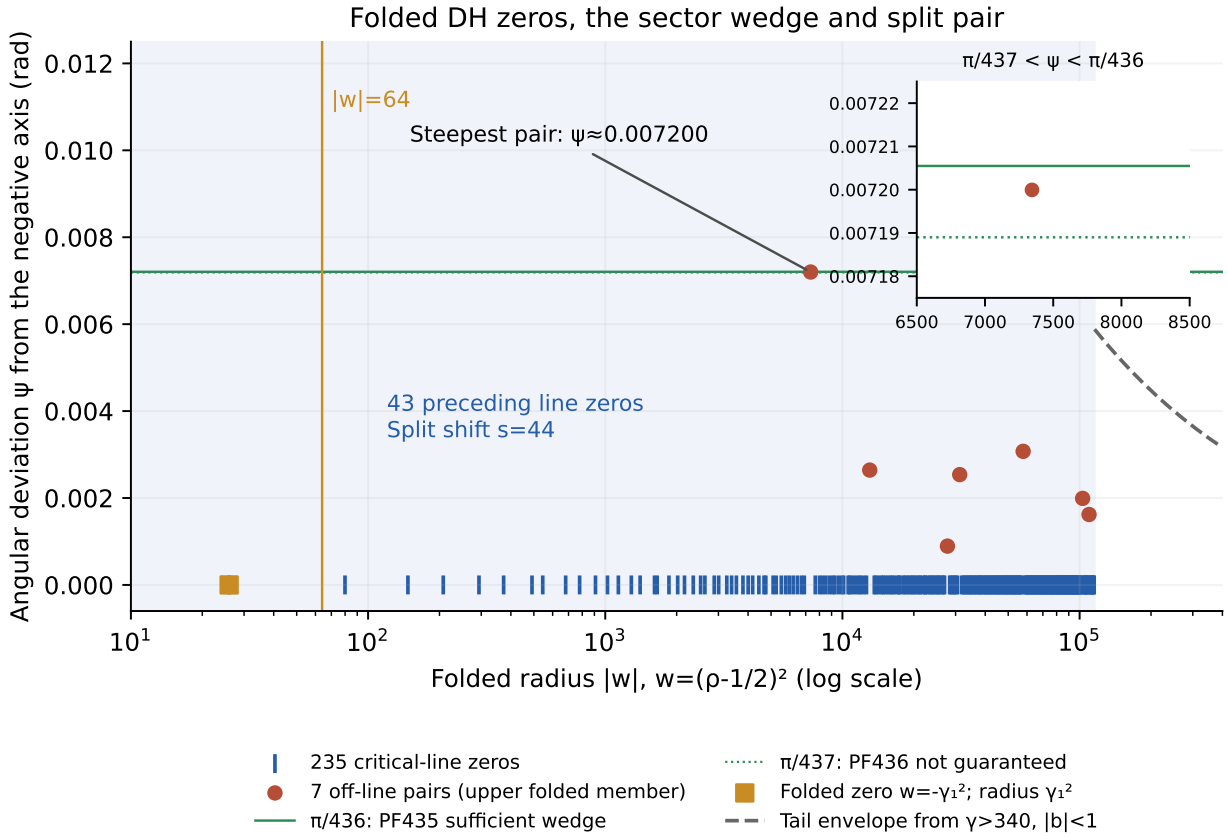


Figure 1: The folded zeros $w = (\rho - \frac{1}{2})^2$ of the classical Davenport–Heilbronn completion on log-radial axes. The horizontal axis is $|w|$; the vertical axis is the angular deviation ψ from the negative real axis. The scales differ by orders of magnitude (the disk $|w| \le 64$, the steepest pair at $|w| \approx 7344$, a 0.41° wedge), so linear polar axes would hide them. Critical-line zeros lie at $\psi = 0$. The steepest off-line zero, $\psi = 0.0071999\ldots$, lies between $\pi/437$ and $\pi/436$ (inset). It therefore sits inside the Schoenberg wedge that gives PF_{435} (Theorem 4.1) but not the one for PF_{436}. For $|w| > 340^2 + 1$, the dashed envelope is $\psi \le 2\arctan((|w| - 1)^{-1/2})$, using $|b| < 1$. Pale shading marks the radius range $|w| \le 340^2$ covered by the atlas. Inside $|w| \le 64$, the only folded zero is $w = -\gamma_1^2$. The corresponding positive pole $a = \gamma_1^2$ of $1/F(-z)$ drives the eventual shift-one positivity (Proposition 5.2). Writing w_0 for the steepest folded pair, the 43 critical-line zeros with $|w| < |w_0|$ place the split pair at shift 44, where the targeted column fails. Data: the certified atlas of Theorem 3.2 (ball midpoints).

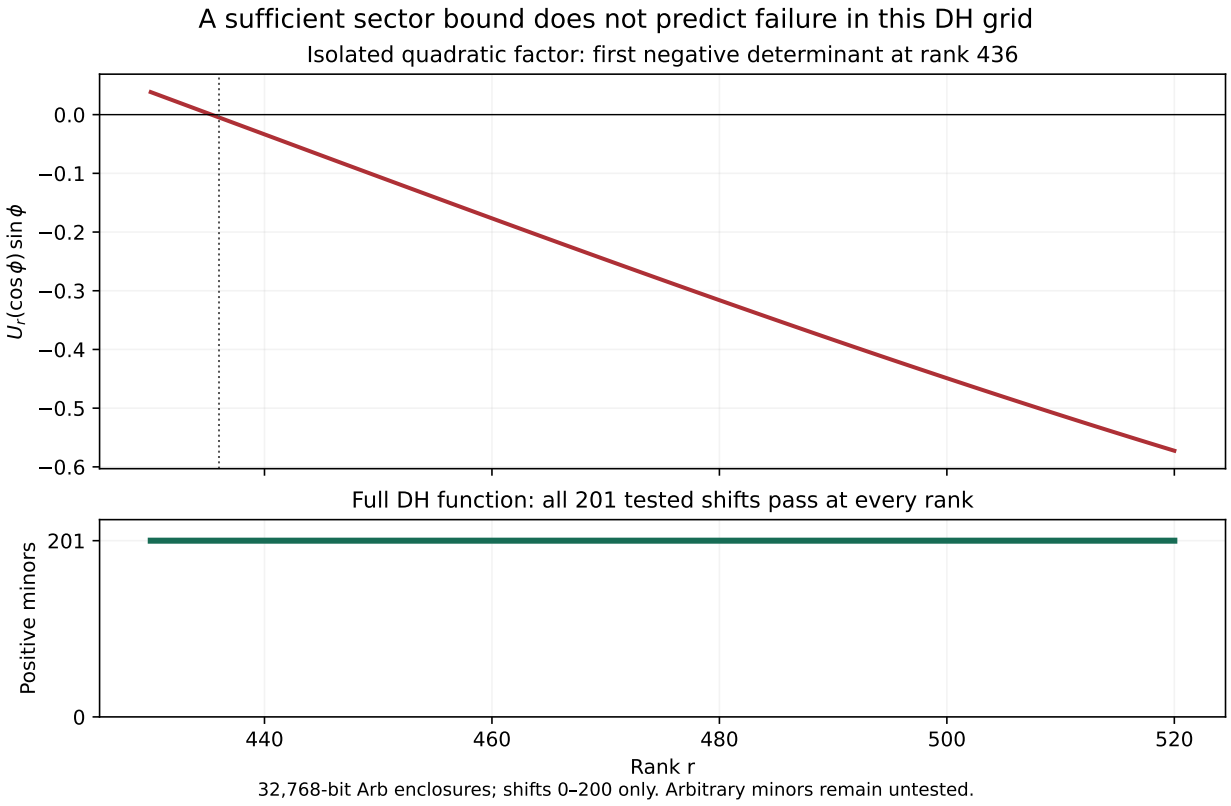


Figure 2: The isolated quadratic determinant, scaled by $\sin\phi$, changes sign at rank 436. Every one of the 201 tested full-function shifts remains positive at each rank through 520. The lower panel counts signs; it does not compare determinant magnitudes or assert positivity of arbitrary minors.

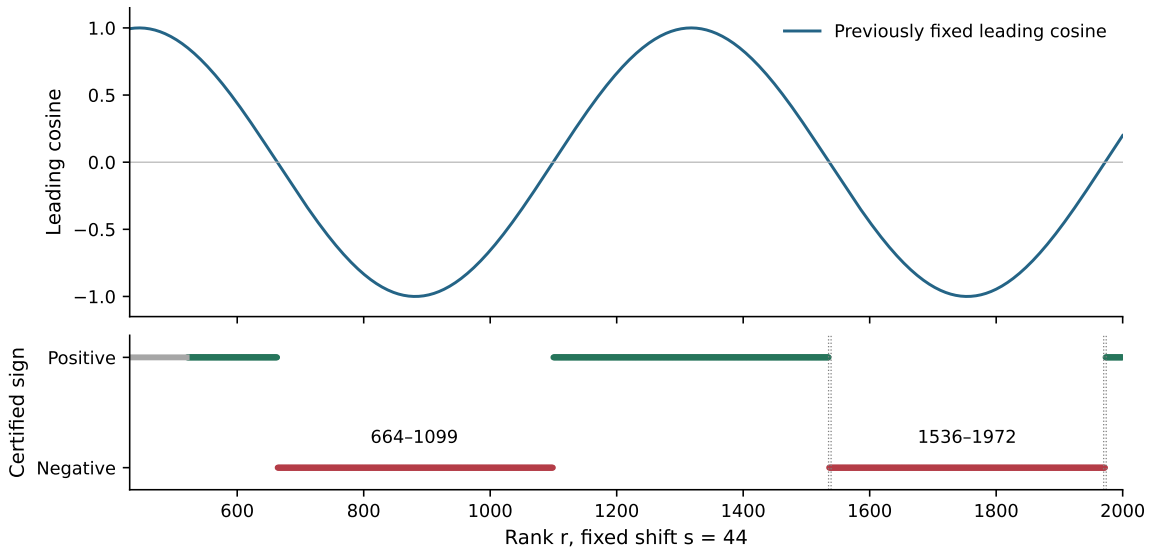


Figure 3: The targeted DH shift-44 column, with certified negative bands 664–1099 and 1536–1972. Dotted lines delimit the pre-recorded second-band entry and last-negative intervals. The first band uses evaluated ordinary coefficients; the second also uses rigorous positive-coefficient tail bounds. The previously fixed leading cosine is an asymptotic diagnostic, with no finite-rank remainder bound.

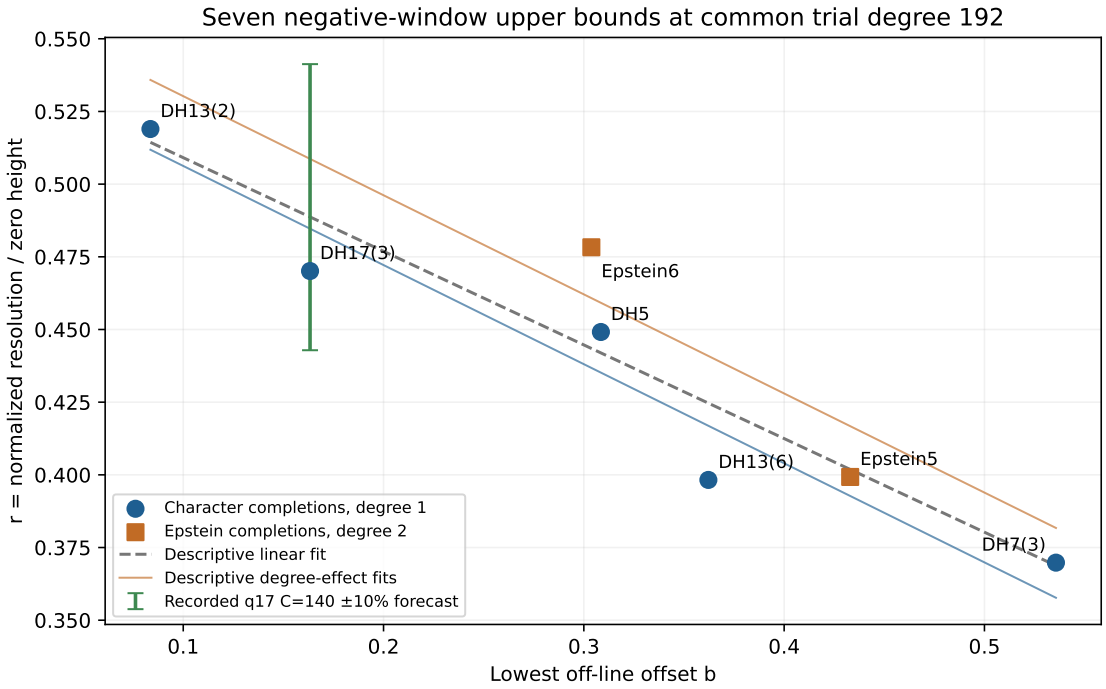


Figure 4: Normalized negative-window upper bounds $r = T_*(C_-)/\gamma$ against the lowest off-line offset b, using common trial degree 192. The fits are descriptive. They do not provide confidence intervals for unknown minimum supports or a zero-free region for zeta. The conductor-17 prediction was supplied before the new window trials.

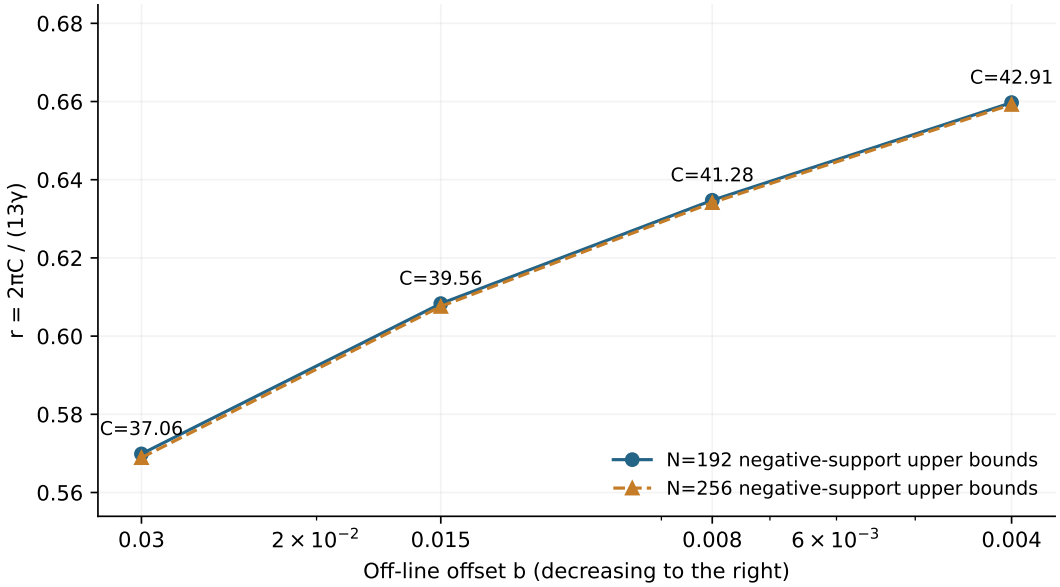


Figure 5: Measured negative-support upper bounds on one rational mixture path, with the off-line offset decreasing to the right. The connecting lines are guides, not fitted limiting laws. The two retained degrees have close endpoints over this tested range. Neither curve supplies a full-support lower bound or determines the $b \to 0$ limit.

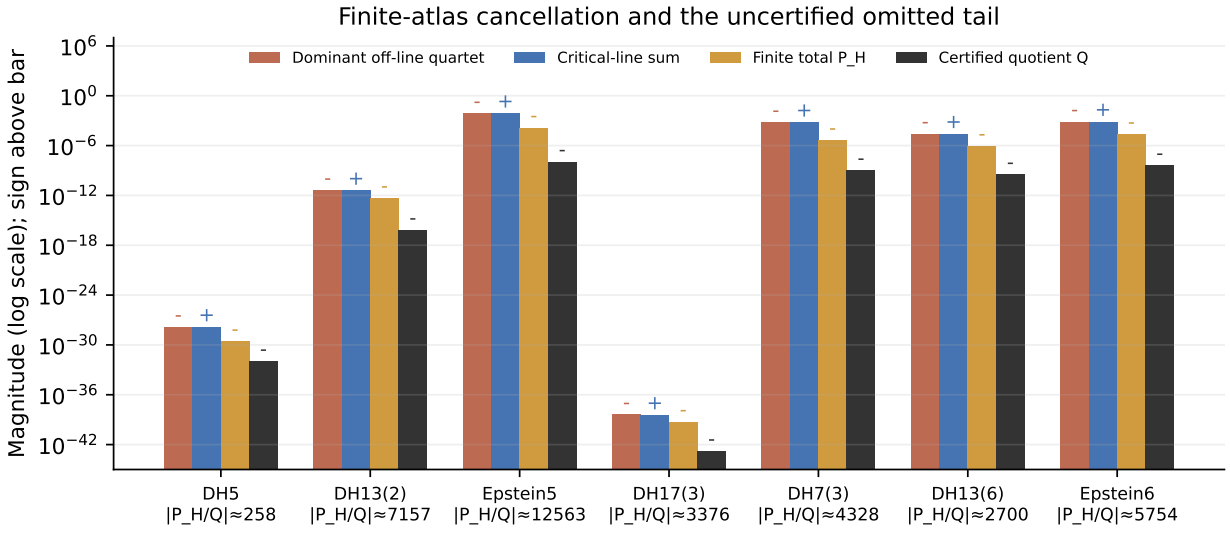


Figure 6: The attribution records of Table 4 on a logarithmic scale; signs are printed above the bars. In every control the dominant off-line quartet (negative) and the critical-line sum over the atlas (positive) have a relative magnitude mismatch below 16%, measured against the critical-line sum (below 3.5% in five of the seven) and largely cancel. The finite total P_H is negative, yet it exceeds the certified quotient Q in magnitude by factors from 258 (DH5) to 12 563 (Epstein5). Under the written explicit formula, $Q - P_H$ is the omitted zero sum above the atlas height; it is not independently enclosed here. The finite sum is therefore mechanism evidence, not a spectral certificate.